# PAVAN KUMAR GUDLA

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## PROFESSIONAL SUMMARY

Applied AI/ML Product Engineer with 2+ years of software engineering experience building enterprise GenAI applications and agentic workflows. Experienced in Retrieval-Augmented Generation (RAG), hybrid retrieval, knowledge graphs, and LLM orchestration using Python, LangGraph, LangChain, and Neo4j. Contributed to AI-powered application engineering workflows spanning specification analysis, frontend code generation, API integration, and LLM observability. Currently pursuing an M. Tech in Artificial Intelligence and Machine Learning at BITS Pilani.

## WORK EXPERIENCE

**Applied AI/ML Product Engineer – Agentic AI & GenAI** | Tata Consultancy Services, Hyderabad | *Mar 2024 – Present*

- Translated evolving business requirements into AI solution designs and working GenAI/Agentic AI workflows, collaborating with product and engineering stakeholders through iterative development, testing, and product integration while emphasizing validation, traceability, and workflow observability.
- Designed and developed multi-agent workflows for an enterprise application-generation platform, transforming business and UI specifications into structured application models, UI previews, and generated code, with a reported reduction of approximately 90% in developer effort.
- Developed GenAI and Agentic AI PoCs for enterprise application engineering, translating evolving requirements into technical designs and working AI workflows through iterative experimentation, testing, and integration.
- Built a Context Analyzer Agent that extracts intent, requirements, screens, fields, widgets, events, and actions from structured and unstructured inputs (BRDs, wireframes, images, HTML), replacing manual requirement analysis 100%.
- Integrated AI agents with Neo4j through API-based JSON model population, storing structured application specifications and meta-model data for retrieval by downstream agents during application-generation workflows.
- Developed Screen Preview and Code Generator Agents that retrieve specs from Neo4j to generate HTML previews and frontend code, shipped in TCS MasterCraft v9.0.0 production release.
- Built and integrated Retrieval-Augmented Generation (RAG) pipelines using text-embedding models, document vectorization, vector search, semantic search and multiple retrieval methods to provide relevant document context to LLM-powered agent workflows.
- Implemented hybrid retrieval capabilities combining embedding-based semantic search and BM25 keyword retrieval, alongside Neo4j Knowledge Graphs and Cypher queries, to retrieve relevant information from document collections and structured enterprise application data.
- Developed reusable Python tools for document and HTML parsing, JSON generation, knowledge retrieval, validation, image preprocessing, and code generation. Integrated internal tools and services through the Model Context Protocol (MCP).
- Integrated enterprise LLMs, including GPT-4.1 Mini and GPT-OSS 120B, and configured system prompts, temperature, top-k, and top-p parameters to support structured, context-grounded generation across AI workflows.
- Implemented LLM workflow observability using Opik to track token consumption, model interactions, agent execution, and workflow-level performance. Contributed to agent architecture design, debugging, and unit, integration, and performance testing within an Agile development lifecycle.

## EDUCATION

**M. Tech in Artificial Intelligence and Machine Learning (WILP)** | Birla Institute of Technology and Science, Pilani (BITS Pilani) | Oct 2025 – Sep 2027

**B. Tech in Mechanical Engineering** | Jawaharlal Nehru Technological University, Kakinada (JNTUK) | Aug 2019 – Jun 2023

## TECHNICAL SKILLS

**Programming & Software Engineering:**

Python • FastAPI • REST APIs • Git • API Integration • Debugging • Unit Testing • Integration Testing • Validation Pipelines

**Frontend & Application Development**:

HTML • Next.js • React • TypeScript • Streamlit

**AI/ML & Generative AI:**

Generative AI • Large Language Models (LLMs) • Agentic AI • Multi-Agent Systems • Agent Orchestration • Tool Calling • Function Calling • Structured Outputs • Prompt Engineering • Context Engineering

**LLM Frameworks & Platforms:**

LangGraph • LangChain • CrewAI • Azure OpenAI • OpenAI • OpenRouter • MCP (Model Context Protocol)

**RAG & Knowledge Systems:**

Retrieval-Augmented Generation (RAG) • Text Embeddings • Semantic Search • Vector Search • ChromaDB • pgvector • Hybrid Retrieval • BM25 • Neo4j • Knowledge Graphs • Cypher • Contextual Grounding

**AI Engineering & Observability:**

LLM Observability • Agent Tracing • Token Usage Monitoring • Opik • Model Evaluation • Output Validation • Human-in-the-Loop (HITL)

**Machine Learning Fundamentals:**

Machine Learning • Deep Learning • Natural Language Processing (NLP) • Recommendation Systems • Probability & Statistics • Linear Algebra

**Cloud, Deployment & DevOps:**

Microsoft Azure • AWS • Docker • Kubernetes • CI/CD • Vercel

## PROJECTS

**IncidentPilot AI | Multi-Agent Incident Analysis Platform**

- Built an AI-powered incident analysis workflow using Python and LangGraph, with specialized triage, diagnosis, and resolution stages.
- Developed a FastAPI backend and Next.js frontend to connect incident inputs with structured analysis results through an API.
- Added automated tests for backend health and analysis workflows; persistence and live agent streaming remain planned for later iterations.

**RecoMart | Recommendation & Data Management Pipeline**

- Built an end-to-end recommendation pipeline over 100,836 ratings and 9,742 movies, combining ratings data with movie and TMDB metadata.
- Implemented data ingestion, validation, cleaning, feature engineering, dataset construction, model training, and recommendation generation.
- Constructed collaborative-filtering, content-based, and hybrid datasets; trained an SVD collaborative-filtering model and evaluated results using RMSE and MAE.

**Autonomous Research Scout | Research Intelligence Platform**

- Built a research discovery and analysis workflow gathering AI research from arXiv, GitHub, and Papers with Code.
- Designed retrieval, LLM-based analysis, prioritization, trend detection, and personalized research recommendation stages.
- Developed and deployed v1 an interactive application presenting AI-generated summaries, research trends, repositories, and personalized reading plans on Streamlit.

**FraudGuard AI | PyTorch-Based Fraud Detection & AWS ML Serving**

- Building a deep-learning fraud detection system using PyTorch, comparing Multilayer Perceptron (MLP) and autoencoder models against conventional machine learning baselines on an imbalanced transaction dataset.
- Developing an ML evaluation pipeline using PR-AUC, precision, recall, F1-score, and cost-sensitive threshold selection to assess fraud detection performance.
- Designing a cloud-native inference architecture using FastAPI, Docker, Amazon S3, Amazon ECR, and Amazon ECS Fargate, with CI/CD automation and CloudWatch monitoring.

## CERTIFICATIONS

- Claude Certified Architect Professional
- Microsoft Certified: Azure AI Engineer Associate
- Microsoft Certified: Azure Data Scientist Associate
- Neo4j Certified Professional